Quantum Computing Final Project

Interim report due: Nov. 8, 2026, 11.59 pm
Final report due: Sunday, Dec. 6, 2026, 11.59 pm.

Submission Policy: Read all the instructions below carefully before you start working on the project and before you make a submission. Submissions should be in teams.

- Read the Academic Honesty Policy. By submitting this project, you affirm that the work is entirely your own.
- If you violate the academic dishonesty policy, then your score for this project will be -MAX = -200.

The goal of this project is for you to explore some important papers in quantum computing and algorithms. The project has two components (each carries 100 points). The first component is to choose one of the papers (according to the rule mentioned below) and investigate it. This should give insight into the results, their detailed analysis and applications, and how they were developed and used. This carries 100 points. The second component is to identify a problem left open or not addressed by the paper and develop an approach to solve it. This part will be evaluated on the strength of your results, with full 100 points for a result that is new and potentially publishable.

List of Papers:

1. Christoph Dürr, Mark Heiligman, Peter Høyer, Mehdi Mhalla: Quantum Query Complexity of Some Graph Problems. SIAM J. Comput. 35(6): 1310-1328 (2006).
2. Harry Buhrman, Christoph Dürr, Mark Heiligman, Peter Høyer, Frédéric Magniez, Miklos Santha, Ronald de Wolf: Quantum Algorithms for Element Distinctness. SIAM J. Comput. 34(6): 1324-1330 (2005)
3. Aram W. Harrow, Avinatan Hassidim, Seth Lloyd. Quantum Algorithm for Solving Linear System of Equations. https://arxiv.org/abs/0811.3171
4. R. Beals et al. Efficient Distributed Quantum Computing. https://royalsocietypublishing.org/rspa/article/469/2153/20120686/83793/Efficient-distributed-quantum-computingEfficient
5. Scott Aaronson, Andris Ambainis: Forrelation: A Problem that Optimally Separates Quantum from Classical Computing. STOC 2015: 307-316.
6. Scott Aaronson, Andris Ambainis: Quantum Search of Spatial Regions. Theory Comput. 1(1): 47-79 (2005)
7. Michael Elkin, Hartmut Klauck, Danupon Nanongkai, Gopal Pandurangan: Can quantum communication speed up distributed computation? PODC 2014: 166-175. Full paper at: https://arxiv.org/abs/1207.5211
8. Gilles Brassard, Peter Hoyer, Michele Mosca, Alain Tapp. Quantum Amplitude Amplification and Estimation. https://arxiv.org/abs/quant-ph/0005055v1
9. Frédéric Magniez, Ashwin Nayak, Jérémie Roland, Miklos Santha: Search via Quantum Walk. SIAM J. Comput. 40(1): 142-164 (2011)
10. Ashwin Nayak, Felix Wu. The quantum query complexity of approximating the median and related statistics. STOC 1999: 384-393. Full paper at: https://arxiv.org/abs/quant-ph/9804066v2
11. François Le Gall, Frédéric Magniez: Sublinear-Time Quantum Computation of the Diameter in CONGEST Networks. PODC 2018: 337-346. Full paper: https://arxiv.org/abs/1804.02917

Your project has to study in some depth one of the above papers. To choose your paper, consider your 7-digit UH ID: sum all the digits of your ID, then take the remainder when you divide the sum by 11. Then add one to it to get the number on the above list. For example, if your UH ID is 7123451, then the sum is 7+1+2+3+4+5+1 = 23 and 23 mod 11 is 1. Adding 1 to it gives 2. So your project will be the 2nd on the list.

For the first component, your study should explain the motivation behind the result, the problem it is solving, its description in clear language, correctness, and mathematical analysis in understandable terms in your *own words* (note that any verbatim use of text or code from other sources such as AI tools will be considered plagiarism and will be subjected to the academic honesty policy). *Note that it is important not only to study the given paper, but also related papers (see the paper's bibliography) to understand the state of the art, including any subsequent papers that might have improved this paper's results.* For the second component, you will identify a specific non-trivial theoretical problem (by non-trivial, meaning that it cannot be obtained easily from the paper itself or other known results) and develop an approach to solve it. Note that obviously, if the open problem or the problem that you are addressing has already been solved by another paper, then this does not count. Usually, the exact problem you are considering would not have been solved (this is like playing the same chess game again, which is rare). **Your solution should contain a full statement of theorems and proofs that are fully understood by you and written by you.**

All references/sources (including any AI tools) you use should appear in a bibliography at the end of the report. **In particular, if you used AI, the entire interaction starting with the prompt should be included.** If AI was used and not disclosed, this will lead to -MAX points. The report should have a maximum length of 20 pages (excluding the title page and the bibliography) written in a 11-pt font on standard A4 paper with 1-in margins on all four sides. The first page should include a title and abstract that summarizes the paper.

Your project will be evaluated on the clarity of the presentation, the originality of the description, and the technical details. You will also give a talk on your work in class (timings, dates, and details will be announced later). In the talk, you will be asked technical questions on your project.

It is important to get started **early** and not wait till the last week.

You should submit an interim report by November 8, 11.59 pm. This will detail the work that you have done till then and outline the remaining work. Most importantly, by this time you should have finished the first component and list the questions that you are pursuing for the second component. You will get feedback on the interim report.

You should submit the final project report in teams (in pdf).